Zero-Temperature SK Full Support: Analytic Dependencies, Rational Coercivity, and Quantitative Stieltjes-Mass Certificates

Dr. Muralidhara Subbarao
Technical Report Assisted by AI Models
www.integralresearch.net
Stony Brook University
Technical Report TR-Ram-OpenAI-722-M565

October 7, 2026
Confidential technical study; verification date October 8, 2026 UTC

Abstract

We examine the pinned manuscript on full support of the zero-temperature, zero-field Sherrington–Kirkpatrick order parameter in the normalization $\xi(t) = t^2/2$. The audit separates compact-strip Parisi regularity, admissible first variations, shape propagation, rational positivity, spatial integration by parts, and the three distinct gap endpoints. A rational modification of the manuscript's finite certificate retains its shape hypotheses and proves an integrated coercivity constant at least $1/500$; the exact optimum in one explicitly restricted coefficient cone is also recorded. A quantitative plateau deficit requires no positive interior lower bound on the gradient's derivative moment. The principal analytic extension is a distributional curvature identity for every admissible monotone integrable coefficient, including jumps, which yields quantitative lower bounds on interior Stieltjes mass from imperfect endpoint contacts. Score, boundary, algebraic, and moment errors enter explicit budgets. A sharp $1/2$ value-stability bound improves the source's $3/2$ control comparison. These are local reformulations and estimates using classical analysis and precise private Ram moment representations, not an independent verification or a new discovery of the full-support theorem. Contribution assessment: C3/5, proved jump-sensitive mass and residual certificates give a significant local extension together with exact coercivity and sharp stability improvements while the full-support theorem retains its analytic dependencies and no global novelty is claimed.

Contents

1 Pinned problem and exact conventions **2**

2 Analytic dependency audit **3**
2.1 Compact-strip construction and stability 3
2.2 Backward shapes and their boundaries 4
2.3 Forward density, score, and ratio 5
2.4 Tails and spatial integration by parts 5
2.5 Curvature and transfer without score convergence 6
2.6 Audit status and the source's later consequences 6

3 First variations and the three gap endpoints **6**
3.1 Every admissible direction, not only atomic ones 6
3.2 Interior and initial gaps 7
3.3 Terminal gap: a separate boundary layer 7

4 Exact improved integrated coercivity **8**
4.1 A rational witness with positive remainder 8

5 Curvature as a measure and quantitative mass **10**
5.1 A plateau deficit without an interior moment minimum 12

6 Error-stable integration and finite moment certificates **13**
6.1 Score defects and boundary flux 13
6.2 Coefficient errors as moment errors 13
6.3 Ram analysis and synthesis on the actual finite space 14

7 Sharp value stability **14**

8 Consequences and remaining boundaries **15**
8.1 Smooth minimizers and the improved density budget 15
8.2 Martingale values and scalar Gaussian approximation 16
8.3 External algorithm and literature comparison 16

9 Verification, advantages, and future work **17**

A Exact optimum in a restricted coefficient cone **17**

B Exact read scope and private correspondences **18**

1 Pinned problem and exact conventions

The object audited is OpenAI, *Full support of the zero-temperature Sherrington–Kirkpatrick order parameter*, September 27, 2026, catalogue entry M565 at commit [1]

`adc7f1241b42e322a6451854ab7e4b4c146bf78a`.

The catalogue's family heading concerns QAOA, but this entry's actual problem is the Parisi support statement; no QAOA result is inferred here. The source consists of eleven TeX files, all read, and a bibliography. Precise read scope is listed in Appendix B. The local source snapshot has no Git metadata, so the catalogue supplies the commit pin; individual input hashes supply the local byte-level provenance.

For independent $J_{ij} \sim N(0,1)$ and $\sigma \in \{-1,1\}^n$, put

$$H_n(\sigma) = n^{-1/2} \sum_{i<j} J_{ij}\sigma_i\sigma_j.$$

With overlap $R_{12} = n^{-1}\sum_i \sigma_i^1\sigma_i^2$, its covariance is $nR_{12}^2/2 - 1/2$. Adding one common independent Gaussian of variance $1/2$ gives covariance $n\xi(R_{12})$, $\xi(t) = t^2/2$, without changing any optimization gap or the expected maximum. In particular $\xi'' = 1$; none of the formulas below silently uses the alternative $\xi(t) = t^2$.

The admissible class is

$$\mathcal{U} = \{\gamma : [0,1) \to [0,\infty) : \gamma \text{ is nondecreasing, right-continuous, and } \|\gamma\|_{L^1(0,1)} < \infty\}. \tag{1.1}$$

All values before one are finite. Define a locally finite positive Borel measure by $\mu_\gamma([0,t]) = \gamma(t)$, so $\mu_\gamma(\{0\}) = \gamma(0)$ and $\mu_\gamma((a,b]) = \gamma(b) - \gamma(a)$. Tonelli gives

$$\int_0^1 \gamma(t)\mathrm{d}t = \int_{[0,1)} (1-s)\mu_\gamma(\mathrm{d}s). \tag{1.2}$$

There is no probability normalization and no requirement of finite total Stieltjes mass. Support uses the relative topology of $[0,1)$, not of $(0,1)$ or $[0,1]$. An atom artificially placed at one cannot be detected by this variational problem.

On $[0,1] \times \mathbb{R}$ define

$$\Phi_\gamma(t,x) = \sup_{|a_s| \le 1} \mathbb{E}\left[\left|x + B_1 - B_t + \int_t^1 \gamma(s)a_s\mathrm{d}s\right| - \frac{1}{2}\int_t^1 \gamma(s)a_s^2\mathrm{d}s\right], \tag{1.3}$$

$$\Phi_t = -\tfrac{1}{2}(\Phi_{xx} + \gamma\Phi_x^2), \qquad \Phi(1,x) = |x|, \tag{1.4}$$

$$\mathcal{P}(\gamma) = \Phi_\gamma(0,0) - \tfrac{1}{2}\int_0^1 t\gamma(t)\mathrm{d}t. \tag{1.5}$$

The controls are progressively measurable for Brownian increments after t, and B has variance rate one. Equation (1.4) is interpreted by monotone step approximation, or equivalently the displayed control value, with classical spatial derivatives only on compact strips before one. The spatial domain is the entire real line: there is no imposed finite-box Dirichlet or Neumann boundary condition.

Set $u = \Phi_x$ and solve the strong stochastic equation

$$\mathrm{d}X_t = \gamma(t)u(t,X_t)\mathrm{d}t + \mathrm{d}B_t, \qquad X_0 = 0. \tag{1.6}$$

Expectations below are for this diffusion unless another law is named. The pinned assertion quantifies over *every* minimizer in $\mathcal{U}$:

$$\gamma \in \arg\min_{\mathcal{U}} \mathcal{P} \quad \Longrightarrow \quad \operatorname{supp}\mu_\gamma = [0,1), \qquad \mathbb{E}u(t,X_t)^2 = t \quad (0 \le t < 1). \tag{1.7}$$

Existence and the equality between the variational minimum and the thermodynamic ground-state energy are external Parisi-theory inputs [2]. Our local results do not prove them or the finite-disorder optimization theorem. The measure identity at zero, the cusp at time one, and the order of all limiting operations are retained throughout.

2 Analytic dependency audit

2.1 Compact-strip construction and stability

Source `regularity.tex`, Lemmas `lem:regularity` and `lem:diffusion`, constructs the needed objects without assuming full support. For a constant c on $[a,b)$, the backward operation is

$$\Phi(t) = c^{-1}\log P_{b-t}e^{c\Phi(b)} \quad (c>0), \qquad \Phi(t) = P_{b-t}\Phi(b) \quad (c=0), \tag{2.1}$$

where P_h is Gaussian convolution with variance h. A smooth terminal approximation is $\psi_M(x) = M^{-1}\log(2\cosh Mx)$, with $0 \le \psi_M - |x| \le M^{-1}\log 2$ uniformly. Evenness, convexity and $|\Phi_x| \le 1$ are preserved. Completion of the square along a controlled diffusion proves (1.3). In particular

$$|x| \le \Phi_\gamma(t,x) \le |x| + \sqrt{2(1-t)/\pi} + \tfrac{1}{2}\int_t^1 \gamma(s)\mathrm{d}s. \tag{2.2}$$

This is uniform in space and is the actual terminal estimate used later.

For $b < 1$, monotonicity gives $\gamma(t) \le \|\gamma\|_1/(1-b)$ on $[0,b]$. Writing $U = \Phi_x$, the mild Burgers equation and $\|\partial_x P_h f\|_\infty \le Ch^{-1/2}\|f\|_\infty$ give

$$\|U_x(t)\|_\infty \le C(b-t)^{-1/2} + C\Gamma \int_t^b (s-t)^{-1/2}\|U_x(s)\|_\infty \mathrm{d}s.$$

The convolution of $h^{-1/2}$ with itself is π, reducing this to an integrable-forcing Gronwall inequality. Higher spatial derivatives are bounded inductively on successively smaller strips; the highest derivative in the product is linear with bounded coefficient U. The proof requires the larger-strip buffer and does not yield constants uniform as $b \uparrow 1$.

The source first obtains value convergence with constant $3/2$, then uses

$$\|F'\|_\infty \le 2\|F\|_\infty/h + (h/2)\|F''\|_\infty$$

successively to obtain uniform convergence of every spatial derivative on $[0,T] \times \mathbb{R}$, $T < 1$. This interpolation applies to the bounded *difference* of two linearly growing value functions. The limiting time equation holds almost everywhere and in integrated form; no classical time derivative at jumps is needed. This is a coherent compact-strip argument. Section 7 sharpens its initial value constant without changing the smoothing proof.

The drift in (1.6) is locally uniformly spatially Lipschitz and bounded in absolute value by $\gamma(t)$. Compact-strip Picard solutions therefore patch uniquely and extend continuously to one. Coupling all approximants with one Brownian path gives a deterministic Gronwall bound for their path differences on compact strips. Passing the smooth Itô identities by isometry yields

$$u(t, X_t) = \int_0^t \Phi_{xx}(s, X_s)\mathrm{d}B_s, \qquad q(t) := \mathbb{E}u(t, X_t)^2, \qquad q'(t) = \mathbb{E}\Phi_{xx}(t, X_t)^2. \tag{2.3}$$

Thus $q \in C^1[0,1)$ and $q(0) = 0$. The initial value zero uses both zero field and $X_0 = 0$. Derivative convergence must be combined with one more bounded spatial derivative when evaluations along coupled paths are compared.

2.2 Backward shapes and their boundaries

Source `signs.tex` starts with $0 < c_1 \le \cdots \le c_N \le M$. On a step with coefficient c, let

$$l = c\Phi, \quad r = l_x, \quad v = r_x, \quad w = r_{xx}, \quad z = r_{xxx}.$$

In backward time τ, $r_\tau = \frac{1}{2}r_{xx} + rr_x$. Its second derivative obeys $w_\tau = \frac{1}{2}w_{xx} + rw_x + 3vw$. Gaussian tilting in (2.1) gives $v > 0$. On $x > 0$, comparison gives

$$w \le 0, \qquad H := w/(2r) + v \le 0, \qquad H_x \ge 0. \tag{2.4}$$

The terminal check is $H = (cM - M^2)\operatorname{sech}^2(Mx) \le 0$; the condition $M \ge \max c_i$ is essential. The two evolution equations are

$$\begin{aligned} H_\tau &= \tfrac{1}{2}H_{xx} + (v/r + r)H_x + vH, \\ (H_x)_\tau &= \tfrac{1}{2}(H_x)_{xx} + (v/r + r)(H_x)_x + ((v/r)_x + 2v)H_x + wH. \end{aligned}$$

At zero, $r = v(0)x + O(x^3)$, so H is smooth even and $v/r + r = x^{-1} + O(x)$. The H comparison is a smooth radial lift in three dimensions, not a one-dimensional comparison with an ignored singular endpoint. For H_x, compare on $[\varepsilon, A]$; the zeroth-order coefficient is bounded above by $2\sup v$ because $(v/r)_x = w/r - v^2/r^2 < 0$. Boundary errors are $O(\varepsilon)$ and $o_A(1)$. A backward jump with ratio $\alpha = c_-/c_+ \le 1$ sends H to $H - (1-\alpha)v$ and H_x to $H_x - (1-\alpha)w$. Both signs survive. These checks identify where monotonicity is used.

2.3 Forward density, score, and ratio

For positive times the diffusion density factors as $f = e^l g$ with $g_t = \frac{1}{2} g_{xx}$ on a step, since its transition kernel is $p_{t-a}(x-y)e^{l(t,x)-l(a,y)}$. Put $s = -g_x/g$ and $a_s = s_x$. The first step has $s = x/t$. At a forward coefficient jump $\delta \geq 0$,

$$g_+ = g_- e^{-\delta\Phi}, \quad s_+ = s_- + \delta\Phi_x, \quad (s_x)_+ = (s_x)_- + \delta\Phi_{xx}.$$

The differentiated Burgers equations give $s_x \geq 1/t$ and $s_{xx} \leq 0$ on $x > 0$. To prove the additional ratio sign, set $d_s = s_x/s$, $d_r = r_x/r$, and $\rho = d_s - d_r$. Then

$$\rho_t = \tfrac{1}{2}\rho_{xx} + d_s\rho_x + (d_r)_x\rho + 2H_x - w - s_{xx}. \tag{2.5}$$

The source is nonnegative; $(d_r)_x < 0$ is an admissible upper zeroth-order bound. Near zero both logarithmic derivatives are $x^{-1} + O(x)$, hence $\rho = O(x)$; at infinity $\rho \to 0$. Comparison on $[\varepsilon, A]$ proves $\rho \geq 0$. Thus

$$j = s_x - (s/r)v = s\rho \geq 0.$$

At a jump $r_+ = \alpha r_-$ and $s_+ = s_- + \lambda r_-$; direct cancellation, not continuity of either score separately, gives $j_+ = j_-$.

2.4 Tails and spatial integration by parts

The source's Lemma `tail:class` proves, with differentiated errors and constants depending on the fixed finite-step data,

$$\begin{aligned} l(t,x) &= cx + C(t) + O(e^{-\eta x}), \\ g(t,x) &= e^{-x^2/(2t)+\beta(t)x+C_0(t)}(1 + O(e^{-\eta x})) \quad (x \to +\infty). \end{aligned}$$

The proof closes these classes under Gaussian tilting and jump multiplication. Evenness controls the negative tails. It follows that $f \leq C\exp(-x^2/(2t) + C|x|)$, r is bounded, $K = rs = O(1 + |x|)$, and all polynomials used in the certificate have polynomial growth. The score estimates are established only away from $t = 0$; using a density at zero would be wrong, since X_0 is deterministic.

Let

$$\begin{gathered} R = r^2, \quad K = rs, \quad b = -w/r, \quad B = b - 2v, \\ S = z + bv - 2Rb = 2rH_x, \quad L_s = -rs_{xx}, \quad \mathcal{D} = r\partial_x. \end{gathered} \tag{2.6}$$

All of R, K, v, B, j, S, L_s are nonnegative even functions. Ratios at zero are smooth extensions; specifically $b(0) = -z(0)/v(0)$, $j(0) = S(0) = 0$. The derivative rules are

$$\mathcal{D}(R, K, v, b, j) = (2Rv,\ 2vK + Rj,\ -Rb,\ -z - bv,\ -jv + Kb - L_s). \tag{2.7}$$

For every polynomial $F(R, K, v, b, j)$,

$$\mathcal{W}(F) := \mathcal{D}F + (v + R - K)F, \qquad f\mathcal{W}(F) = \partial_x(rfF), \qquad \mathbb{E}_f\mathcal{W}(F) = 0. \tag{2.8}$$

Here $f_x/f = r - s$. Absolute integrability and $rfF \to 0$ at both infinities follow from the preceding tails. The polynomial is F_2 in the source, not the square of an unspecified F. A cutoff is removed only after its boundary term is controlled.

2.5 Curvature and transfer without score convergence

On a positive plateau $\gamma = c$, along X the drifts of (r, v, w) are $(0, -v^2, -3vw)$ and their diffusion coefficients are (v, w, z). Consequently

$$q'' = c^{-2}\mathbb{E}(w^2 - 2v^3), \qquad q''' = c^{-2}\mathbb{E}(z^2 - 12vw^2 + 6v^4). \tag{2.9}$$

The coefficient -12 consists of -6 from $\mathrm{d}(w^2)$ and -6 from $-2\mathrm{d}(v^3)$; the diffusion correction in $\mathrm{d}(v^3)$ is not optional. The fourth-derivative term is $z = c\Phi_{xxxx}$.

Positive monotone step approximants can retain a chosen positive plateau exactly while converging in L^1. Uniform spatial derivatives through order four, and the next derivative for path evaluation, transfer the integrated certificate to the limit. No limit of s, j, b or their tail constants is asserted or needed. Strict positivity of Φ_{xx} on a plateau follows from the tilted variance of the nonconstant terminal-strip gradient. At $c = 0$, division by c is forbidden: the independent heat argument gives $q'' = \mathbb{E}\Phi_{xxx}^2 > 0$. Strictness follows from a positive Gaussian transition density and the impossibility of a one-Lipschitz even quadratic differing boundedly from $|x|$.

2.6 Audit status and the source's later consequences

The foregoing is a stepwise mathematical audit of the source's analytic interfaces, not machine verification of all maximum-principle, stochastic-calculus, and limiting arguments. They are retained as explicit analytic dependencies of the improved certificate. The finite rational identities are a separate, exactly replayed component. No defect in the displayed local curvature identities was found.

The source's final files do not extend the scope of the core proof. `order-parameter-regularity.tex` uses saturated consistency, a continuous moment ratio and simultaneous polynomial-moment bootstrapping. `consequences.tex` proves a bounded martingale terminal spin, a scalar Gaussian Euler approximation, and a shifted coefficient product; it does not implement a finite-disorder algorithm. `algorithm-application.tex` invokes an external theorem with fixed-accuracy, real-arithmetic complexity $C(\varepsilon)n^2$. The source expressly leaves preprocessing and bit complexity outside that count. We keep those boundaries in Section 8.

3 First variations and the three gap endpoints

3.1 Every admissible direction, not only atomic ones

Suppose $\eta \in L^1(0,1)$ and $\gamma + \epsilon\eta \in \mathcal{U}$ for $0 \le \epsilon \le \epsilon_0$. For $\beta = \gamma + \epsilon\eta$ put $D = \Phi_\beta - \Phi_\gamma$ and $\widetilde{u} = \Phi_{\beta,x}$. Subtracting the two PDEs gives exactly

$$D_t + \tfrac{1}{2}D_{xx} + \tfrac{1}{2}\beta(\widetilde{u} + u)D_x = -\tfrac{1}{2}(\beta - \gamma)u^2.$$

The linearizing diffusion Y^ϵ, started at zero, has drift $\beta(\widetilde{u} + u)/2$. Compact-strip approximation justifies Itô with the merely measurable time coefficient. Letting $T \uparrow 1$ removes $\mathbb{E}D(T, Y_T^\epsilon)$ because its absolute value is at most $\frac{3}{2}\int_T^1 |\beta - \gamma|$. Therefore

$$\frac{\Phi_\beta(0,0) - \Phi_\gamma(0,0)}{\epsilon} = \frac{1}{2}\int_0^1 \eta(t)\mathbb{E}u(t, Y_t^\epsilon)^2 \mathrm{d}t.$$

Compact-strip convergence of Y^ϵ to X and domination by $|\eta|$ imply

$$D^+\mathcal{P}(\gamma)[\eta] = \frac{1}{2}\int_0^1 \eta(t)(q(t) - t)\mathrm{d}t. \tag{3.1}$$

This argument handles the terminal cusp through an actual uniform bound; differentiating a smooth-terminal formula at one is not a substitute.

For a minimizer set $G(s) = \int_s^1 (q(t) - t)\mathrm{d}t$. Adding an atom at any $s < 1$ is admissible and gives $G(s) \geq 0$. For compact $K_0 \subset [0, 1)$, removal of $\nu = \mu_\gamma|_{K_0}$ gives the bounded admissible direction $\eta(t) = -\nu([0, t])$. Fubini gives

$$0 \leq D^+\mathcal{P}[\eta] = -\tfrac{1}{2}\int_{K_0} G(s)\mu_\gamma(\mathrm{d}s).$$

Hence $G = 0$ on the relative support by continuity. This removal works for singular continuous mass as well as atoms. There is no probability constraint requiring compensating mass elsewhere. At every interior support point s, $G' = s - q(s)$ and $G'' = 1 - q'(s)$ yield $q(s) = s$, $q'(s) \leq 1$. If zero is in the support, $G(0) = G'(0) = 0$ and the one-sided second-order expansion gives $q'(0) \leq 1$. It would be incorrect to assume $q'(s) = 1$ at an isolated contact before full support has been proved.

3.2 Interior and initial gaps

For endpoints a, b in the support with $0 \leq a < b < 1$, an open support gap has constant $\gamma = c$ there. If $c > 0$, the integrated coercivity and (2.9) give $q''' > 0$, so q' is strictly convex and strictly below its endpoint chord. As both endpoint values are at most one, $q(b) - q(a) < b - a$, contradicting contact. The functions q and q' are continuous up to these endpoints even if γ jumps there; q'', represented almost everywhere by J below, can jump. For $c = 0$, strict increase of q' instead gives $q'(t) < q'(b) \leq 1$. For an initial component $[0, b)$ disjoint from support, the convention $\mu(\{0\}) = \gamma(0)$ forces $c = 0$; use $q(0) = 0$ and contact at b. A positive plateau issuing from zero is possible only if an atom at zero is retained, in which case zero belongs to support and the previous endpoint argument applies.

3.3 Terminal gap: a separate boundary layer

Assume a minimizer is constant $c < \infty$ on $(a, 1)$ and choose $s \in (a, 1)$. Put $h(t, y) = \mathbb{E}e^{c|y+\sqrt{1-t}Z|}$ when $c > 0$; for $c = 0$ put $h = 1$. The transition density is

$$k_{s,t}(x, y) = p_{t-s}(y - x)h(t, y)/h(s, x).$$

Choose $R < \infty$ with $\Pr(|X_s| \leq R) = p_0 > 0$, and put $\ell_0 = 1 - s$. For $t \geq s + \ell_0/2$, $|x| \leq R$, $|y| \leq 1$,

$$p_{t-s}(y - x) \geq (2\pi\ell_0)^{-1/2}e^{-(R+1)^2/\ell_0} =: k_* > 0, \qquad h(s, x) \leq e^{cR}\mathbb{E}e^{c\sqrt{\ell_0}|Z|} =: M.$$

As $h(t, y) \geq 1$, the density of X_t is at least $m = p_0k_*/M$ on $[-1, 1]$ for all such t. For $\varepsilon = 1 - t$, $|y| \leq \sqrt{\varepsilon}$, the tilted Gaussian sign formula for $u(t, y)$ gives each sign probability at least $\theta = \Pr(Z \geq 1)/(e^c\mathbb{E}e^{c|Z|}) > 0$. Consequently $1 - u(t, y)^2 \geq 4\theta(1 - \theta)$ and

$$1 - q(t) \geq C\sqrt{1 - t}, \qquad C = 8m\theta(1 - \theta) > 0. \tag{3.2}$$

Then $q(t) - t \leq (1 - t) - C\sqrt{1 - t} < 0$ sufficiently close to one, contradicting $G \geq 0$. The quantifier is uniform over all sufficiently late times for this fixed hypothetical plateau, not over arbitrary coefficients. This excludes empty support as well. Compact-strip curvature alone would not exclude this endpoint.

4 Exact improved integrated coercivity

4.1 A rational witness with positive remainder

Retain the signs, derivative rules, and exact density relation in Section 2. Define the rational constants

$$(A, C, k, T_0, g_0, h) = (1/13, 53/24, 19/16, 73/48, 2331/10000, 87/2500)$$

and retain the source polynomials

$$\begin{aligned} Q &= Avb + Cv^2 + kKv - 2kRb + T_0Rv, \qquad T = v(g_0R + hj), \qquad (4.1)\\ P_0 &= Rj(b - 7v)^2, \quad P_1 = RB(b - 7v/3)^2, \quad P_2 = Kv(b - 9v/2)^2,\\ P_3 &= vB(K + R/4 - v/7)^2,\\ P_4 &= (vb - 14v^2/5 - 13Kv/4 + 4Rv/3)^2. \qquad (4.2) \end{aligned}$$

Only the five weights and four coefficients independent of b in the divergence correction are changed. Put

$$\begin{aligned} F_* = (Av - 2kR)b^2 + b\{2Cv^2 + 2kKv + (g_0 + 2T_0)Rv + hjv\}\\ + x_*v^2j + y_*v^3 + m_*Kv^2 + n_*Rv^2, \end{aligned} \qquad (4.3)$$

with

$$\begin{aligned} (p_0, p_1, p_2) &= (21639/625000, 62607/1000000, 52641/1250000),\\ (p_3, p_4) &= (22732413/10000000, 159567/2500000),\\ (x_*, y_*) &= (463697/500000, 3424743/5000000),\\ (m_*, n_*) &= (-26249939/10000000, -10734887/5000000). \end{aligned} \qquad (4.4)$$

All displayed numbers are exact rationals. In particular $x_* > 0$.

Theorem 4.1 (Improved rational certificate)**.** *With $B = b - 2v$, $S = z + bv - 2Rb$, and the derivation (2.7), the following is a polynomial identity over $\mathbb{Q}$:*

$$\begin{aligned} z^2 - 12Rvb^2 + 6v^4 = (z + Q)^2 + ST + \sum_{i=0}^{4} p_iP_i + \mathcal{W}(F_*)\\ + L_s(x_*v^2 + hbv) + P_*^{\mathrm{rem}}. \end{aligned} \qquad (4.5)$$

The nonzero coefficients of P_*^{rem} are exactly those in Table 1, in the independent variables R, K, v, B, j. If the analytic hypotheses of (2.8) hold, then

$$\mathbb{E}_f(z^2 - 12vw^2 + 6v^4) \geq \kappa_*\mathbb{E}_fv^4 \geq \frac{1}{500}\mathbb{E}_fv^4, \qquad \kappa_* = \frac{95187319}{47531250000}. \qquad (4.6)$$

Proof. The choice of the first two lines of F_* gives $\partial_bF_* = 2Q + T$. Since $\mathcal{D}b = -z - bv$, all terms linear in z cancel in (4.5); the z^2 terms cancel too. Also $\partial_jF_* = x_*v^2 + hbv$, so the $-L_s$ term of $\mathcal{D}j$ is exactly compensated by the displayed L_s term. After substitution $b = B + 2v$, the residual depends only on R, K, v, B, j. Expansion gives Table 1, with every unlisted coefficient zero. This finite equality is exactly checkable using the displayed derivation and rationals; an exact coefficient replay, including the absent terms, was executed. All coefficients in the table are positive. The prefactors of the five squares are $Rj, RB, Kv, vB, 1$; $S, T, L_s, x_*v^2 + hbv$ are nonnegative. Integrating removes $\mathcal{W}(F_*)$ by (2.8). Finally $Rb^2 = w^2$ and $\kappa_* - 1/500 = 124819/47531250000 > 0$. □

Monomial	Coefficient	Monomial	Coefficient
R^2v^2	2503/180000000	R^2vB	7644457/480000000
R^2B^2	1097/120000	RKv^2	1/93750
$RKvB$	9226993/60000000	Rv^3	21067/2925000000
Rv^2B	36169387/5460000000	Rv^2j	4159/10000000
RvB^2	107/1500000	$RvBj$	3/250000
RB^3	186109/13000000	RB^2j	111/625000
K^2v^2	1627171/40000000	K^2vB	1017587/10000000
Kv^3	3290233/1950000000	Kv^2B	11/1312500
Kv^2j	498497/500000	KvB^2	167/16250000
$KvBj$	87/2500	v^4	95187319/47531250000
v^3B	12768437/1242150000000	v^2B^2	3033177/422500000

Table 1: All 22 nonzero remainder coefficients for the compact rational witness. Strict positivity refers to these coefficients, not to every possible larger polynomial ansatz.

Proposition 4.2 (A coefficient-only perturbation radius)**.** *Keep Q, T, P_i fixed and perturb the nine parameters in (4.4) by a vector Δu satisfying*

$$\|\Delta u\|_\infty \le \rho_* = 11/16781250.$$

Form the divergence correction from the perturbed parameters exactly. All weights and its coefficient x remain nonnegative, every remainder coefficient remains nonnegative, and the v^4 coefficient remains at least $1/500$. Therefore the clean integrated bound is retained under the same analytic hypotheses.

Proof. After the structural cancellations, each coefficient is affine, $r_\alpha(u) = d_\alpha + a_\alpha \cdot u$, on the same 22-monomial set. The possible change is at most $\|a_\alpha\|_1\|\Delta u\|_\infty$. Subtract $1/500$ from the target v^4 coefficient and zero from the others. Exact rational evaluation of all nonconstant rows gives

$$\min_\alpha \frac{r_\alpha(u_*) - \mathbf{1}_{\alpha=v^4}/500}{\|a_\alpha\|_1} = \frac{11}{16781250}.$$

Zero-slope rows already have nonnegative margin and are omitted from the minimum. The minimum is attained by the BKv^2 row; its coefficient is $11/1312500$ and its row norm is $179/14$. The five weights and x exceed ρ_*, so their signs persist. The affine row construction is the coefficient expansion in the proof of Theorem 4.1; no new monomials occur under these nine perturbations. □

This is an algebraic robustness radius, not a tolerance for inaccurate PDE fields, density scores, moment estimates, or violations of the shape inequalities. Those errors are treated in Section 6.

The source retains $1661507/1521000000$ and states the weaker $1/1000$. The present witness therefore more than doubles the *stated* constant, but does not double the actual retained source constant. The ratio to the latter is about 1.8333. Its significance here is an increased residual budget in the quantitative estimates below, not a larger class of spin glasses or a faster algorithm. Appendix A records a slightly larger optimum and a dual certificate only for a fixed finite coefficient cone.

5 Curvature as a measure and quantitative mass

This section extends the plateau calculation to arbitrary admissible γ, before minimality, continuity, or full support is assumed. Use unscaled derivatives to avoid division by a vanishing coefficient:

$$\begin{gathered} A_t = \Phi_{xx}(t, X_t), \quad C_t = \Phi_{xxx}(t, X_t), \quad D_t = \Phi_{xxxx}(t, X_t), \\ M_k(t) = \mathbb{E}A_t^k, \quad N(t) = \mathbb{E}C_t^2, \quad F(t) = q'(t) = M_2(t), \qquad J(t) = N(t) - 2\gamma(t)M_3(t), \\ \mathcal{K}(t) = \mathbb{E}D_t^2 - 12\gamma(t)\mathbb{E}(A_tC_t^2) + 6\gamma(t)^2M_4(t). \end{gathered} \tag{5.1}$$

Here C_t is a derivative, not the rational constant C of Section 4. All moments are finite on compact strips.

Lemma 5.1 (Fixed-time transfer without a support hypothesis)**.** *For every $\gamma \in \mathcal{U}$ and every $t \in (0,1)$,*

$$\mathcal{K}(t) \geq \kappa\gamma(t)^2M_4(t), \qquad \kappa = 1/500. \tag{5.2}$$

The larger κ_* in (4.6) is also admissible.

Proof. If $c = \gamma(t) > 0$, replace γ on a shrinking interval $[t-\delta_n, t+\delta_n)$ by c. Monotonicity is retained because all values to the left are at most c and all values to the right are at least c. The change is L^1 small even at a jump: on a fixed compact strip γ is bounded. Approximate on the two remaining sides by bounded positive monotone steps, preserve the central plateau, and choose $M_n \to \infty$ above their heights. At the fixed time use Theorem 4.1 with $v_n = cA_n$, $w_n = cC_n$, $z_n = cD_n$, divide by c^2, and pass to the limit using only compact-strip derivative and diffusion convergence. This gives the claim for the specified right-continuous value $\gamma(t)$, without assuming γ continuous there. If $c = 0$ then $\mathcal{K}(t) = \mathbb{E}D_t^2 \geq 0$, which is exactly the required bound. □

Theorem 5.2 (Distributional curvature with Stieltjes jumps)**.** *For every $\gamma \in \mathcal{U}$, F is locally absolutely continuous on $[0,1)$, J is right-continuous and locally of bounded variation, and $F' = J$ almost everywhere. On $(0,1)$, as an equality of locally finite signed measures,*

$$\mathrm{d}J = \mathcal{K}(t)\mathrm{d}t - 2M_3(t)\mu_\gamma(\mathrm{d}t), \qquad \mathrm{d}J \geq \kappa\gamma(t)^2M_4(t)\mathrm{d}t - 2M_3(t)\mu_\gamma(\mathrm{d}t). \tag{5.3}$$

In particular at a jump $t > 0$,

$$J(t) - J(t-) = -2M_3(t)(\gamma(t) - \gamma(t-)). \tag{5.4}$$

Proof. For $f_k = \partial_x^k\Phi$, spatial differentiation and Itô give

$$\mathrm{d}f_k(t, X_t) = -\frac{\gamma(t)}{2}\sum_{i=1}^{k-1}\binom{k}{i}f_{i+1}(t, X_t)f_{k-i+1}(t, X_t)\mathrm{d}t + f_{k+1}(t, X_t)\mathrm{d}B_t. \tag{5.5}$$

For $k = 1$ the sum is empty. Finite-step approximation with a common Brownian motion proves this identity for integrable coefficients; every stochastic integrand is bounded on a fixed strip and every drift is dominated by a constant times γ. Thus no time derivative of γ is used in (5.5). For $k = 2, 3$ the drifts are $-\gamma A^2$ and $-3\gamma AC$. Itô's product rule, followed by expectation, yields the almost everywhere identities

$$F' = N - 2\gamma M_3, \tag{5.6}$$

$$N' = \mathbb{E}D_t^2 - 6\gamma\mathbb{E}(A_tC_t^2),$$

$$M_3' = 3\mathbb{E}(A_tC_t^2) - 3\gamma M_4. \tag{5.7}$$

The integral forms show that F, N, M_3 are locally absolutely continuous. Since γ is locally BV, $J = N - 2\gamma M_3$ is BV and right-continuous. The BV product rule with the continuous function M_3 gives

$$\mathrm{d}J = (N' - 2\gamma M_3')\mathrm{d}t - 2M_3\mathrm{d}\gamma.$$

Substitution of (5.7) proves the equality in (5.3). On $(0,1)$, $\mathrm{d}\gamma = \mu_\gamma$; the artificially fixed left value zero at time zero plays no role in this distributional identity. Lemma 5.1 proves the inequality. Continuity of all moments gives (5.4). □

The negative singular term is essential. Deleting it would incorrectly claim global convexity of q' for a nonconstant monotone coefficient. The source proves the smooth-minimizer analogue only after saturated consistency; Theorem 5.2 is a nonsmooth pre-minimality identity. Its ingredients are classical Itô calculus and BV product rules.

Theorem 5.3 (Weighted mass identity and lower certificate)**.** *Fix $0 \le a < b < 1$, write $\ell = b - a$, and set*

$$w_{a,b}(t) = (t-a)(b-t), \qquad D_{a,b} = \frac{\ell}{2}(F(a) + F(b)) - (q(b) - q(a)).$$

Then

$$2\int_{(a,b)} w_{a,b} M_3 \mathrm{d}\mu_\gamma = \int_a^b w_{a,b}\mathcal{K}\mathrm{d}t - 2D_{a,b}. \tag{5.8}$$

Consequently, for any certified $V_3 > 0$ with $\sup_{[a,b]} M_3 \le V_3$,

$$\mu_\gamma((a,b)) \ge \frac{2}{V_3\ell^2}[\kappa\int_a^b w_{a,b}\gamma^2 M_4 \mathrm{d}t - 2D_{a,b}]_+. \tag{5.9}$$

Endpoint atoms are not included and do not enter either weighted integral.

Proof. Integrate (5.3) against $w = w_{a,b}$. The endpoints vanish. Since $F' = J$ almost everywhere, two ordinary BV/AC integrations by parts give

$$\int_{(a,b)} w\mathrm{d}J = -\int_a^b w' J\mathrm{d}t = -[w'F]_a^b + \int_a^b w'' F\mathrm{d}t = \ell(F(a) + F(b)) - 2\int_a^b F\mathrm{d}t = 2D_{a,b}.$$

This holds also for $a = 0$ by right limits; $w(0) = 0$ eliminates any possible endpoint atom. Rearranging proves (5.8). Use (5.2) and $0 \le wM_3 \le \ell^2 V_3/4$ to obtain (5.9). If its numerator is nonpositive, the conclusion is simply the pre-existing nonnegativity of the measure. □

Corollary 5.4 (Imperfect contacts force interior mass)**.** *Suppose on $[a,b]$ that $\gamma \ge c \ge 0$, $M_4 \ge m_4 \ge 0$, $M_3 \le V_3 > 0$, and*

$$F(a), F(b) \le 1 + \epsilon, \qquad q(b) - q(a) \ge \ell - \delta, \qquad \epsilon, \delta \ge 0.$$

Then

$$\mu_\gamma((a,b)) \ge [\frac{\kappa c^2 m_4 \ell}{3V_3} - \frac{4\epsilon}{V_3\ell} - \frac{4\delta}{V_3\ell^2}]_+. \tag{5.10}$$

More generally, if only $\mathcal{K} \ge \kappa\gamma^2 M_4 - e(t)$ is certified with $e \ge 0$ integrable, subtract $2\int_a^b w_{a,b} e\mathrm{d}t/(V_3\ell^2)$ inside the positive part. In particular this extra loss is at most $\|e\|_{L^1(a,b)}/(2V_3)$.

Proof. $D_{a,b} \le \ell\epsilon + \delta$ and $\int_a^b w_{a,b}\mathrm{d}t = \ell^3/6$. Substitute in (5.9). For the residual version retain the $-\int we$ term throughout the proof; use $w \le \ell^2/4$ for the last assertion. □

For a hypothetical positive plateau the left side is zero. The strict inequality

$$\kappa c^2 m_4 \ell^3/6 > 2\ell\epsilon + 2\delta + \int_a^b w_{a,b}e\mathrm{d}t \tag{5.11}$$

is therefore a finite-error exclusion test. In the exact-contact case it recovers the source contradiction but uses a weighted integral, not pointwise convexity testing. It also proves positive mass for nonconstant candidates whose entire interior is not known to be in contact. Thus its use is not confined to literal gaps. The positive lower bound m_4 and the upper bound V_3 are assumptions to certify, not numbers extracted from unverified samples. Jensen gives $M_4 \geq F^2$; a Hessian supremum $A_t \leq H_T$ gives $M_3 \leq H_T^3$ on a strip. When only the four endpoint data are approximate, certified enclosures for those data directly produce ϵ, δ.

5.1 A plateau deficit without an interior moment minimum

The next estimate is useful when certifying m_4 would be unnecessarily expensive. It uses Jensen's inequality and the endpoint chord, and is restricted to a genuine constant interval with the exact curvature inequality. General approximate-curvature errors remain covered by Corollary 5.4, not silently absorbed here.

Theorem 5.5 (Quantitative positive-plateau deficit)**.** *Let $g \in W^{2,1}(a,b)$ have its continuous endpoint values, with $g \geq 0$, $g(a), g(b) \leq A_0$, and $g'' \geq \lambda g^2$ almost everywhere, where $A_0, \lambda \geq 0$ and $\ell = b - a > 0$. Then*

$$D := A_0\ell - \int_a^b g(t)\mathrm{d}t \geq \frac{\lambda A_0^2 \ell^3}{12(1 + \lambda A_0 \ell^2/4)}. \tag{5.12}$$

For a positive plateau $\gamma = c$ in the Parisi problem, take $g = q'$, $\lambda = \kappa c^2$. If certified endpoint errors satisfy

$$|q(a) - a|, |q(b) - b| \leq \epsilon_q, \qquad q'(a), q'(b) \leq 1 + \epsilon_d, \qquad \epsilon_q, \epsilon_d \geq 0,$$

a necessary condition for that plateau is

$$\epsilon_d \ell + 2\epsilon_q \geq \frac{\kappa c^2 (1+\epsilon_d)^2 \ell^3}{12(1 + \kappa c^2(1+\epsilon_d)\ell^2/4)}. \tag{5.13}$$

Proof. Convexity gives $0 \leq g \leq A_0$. Put $d = A_0 - g \geq 0$, so $\int d = D$. The same two integrations used in Theorem 5.3 give

$$D \geq \frac{\ell}{2}(g(a) + g(b)) - \int_a^b g = \frac{1}{2}\int_a^b w_{a,b} g'' \geq \frac{\lambda}{2}\int_a^b w_{a,b} g^2.$$

Since $g^2 = (A_0 - d)^2 \geq A_0^2 - 2A_0 d$, and $\int wd \leq \ell^2 D/4$, this implies

$$D \geq \lambda A_0^2 \ell^3/12 - \lambda A_0 \ell^2 D/4.$$

Rearrange to prove (5.12). On a positive plateau, (2.9), the improved integrated inequality and Jensen give

$$g'' = q''' \geq \kappa c^2 \mathbb{E} A_t^4 \geq \kappa c^2 (\mathbb{E} A_t^2)^2 = \kappa c^2 g^2.$$

All required derivatives are integrable up to $a, b < 1$ by compact-strip bounds and the constant interior coefficient. With $A_0 = 1 + \epsilon_d$, the endpoint contact errors give $D \leq \epsilon_d \ell + 2\epsilon_q$; substitute in (5.12). □

For exact contacts and slopes the left side of (5.13) is zero, while its right side is positive for $c, \ell > 0$. For imperfect contacts it gives an explicit tolerance without assuming $\inf q' > 0$ or estimating the density in space. Improving κ increases the right side monotonically; the derivative of $\kappa/(1 + C\kappa)$ is $(1 + C\kappa)^{-2} > 0$. This makes the finite coefficient gain operational in a residual exclusion test, although no numerical candidate or endpoint error has been certified in this report.

6 Error-stable integration and finite moment certificates

6.1 Score defects and boundary flux

The exact polynomial identity has content only relative to its measure. The following version allows a certified score defect without claiming that an approximate density is the law of the Parisi diffusion.

Theorem 6.1 (Residual and score budget)**.** *Let $f > 0$ be a C^1 probability density on $\mathbb{R}$; retain smooth functions and the definitions, signs, and derivative rules of Section 4. Assume*

$$f_x/f = r - s + e, \qquad e \in L^2(f), \qquad rF_* \in L^2(f).$$

Suppose all terms of (4.5) are integrable, and the boundary difference $\mathcal{B} = \lim_{A\to\infty}[rfF_*]_{-A}^{A}$ exists and obeys $|\mathcal{B}| \le \beta$. If the exact algebraic identity is perturbed by an additive residual E_{alg} with $\mathbb{E}_f|E_{\mathrm{alg}}| \le \eta$, then

$$\mathbb{E}_f(z^2 - 12vw^2 + 6v^4) \ge \kappa_*\mathbb{E}_f v^4 - \beta - \eta - \|rF_*\|_{L^2(f)}\|e\|_{L^2(f)}. \tag{6.1}$$

The algebraic residual convention is left side minus the displayed right side of (4.5).

Proof. The perturbed logarithmic derivative gives $\partial_x(rfF_*) = f\mathcal{W}(F_*) + rF_*ef$. Integrating and taking the indicated limit gives $\mathbb{E}_f\mathcal{W}(F_*) = \mathcal{B} - \mathbb{E}_f(rF_*e)$. Integrate the perturbed identity, discard the nonnegative terms other than $\kappa_* v^4$, and use Cauchy–Schwarz and $E_{\mathrm{alg}} \ge -|E_{\mathrm{alg}}|$. □

For a finite interval $[-A, A]$, the same derivation retains the actual flux $[rfF_*]_{-A}^{A}$. Its integrals are unnormalized unless the mass in the box is explicitly divided out; normalizing changes every moment and error by that same mass. A whole-line conclusion requires a bound on the omitted curvature integral, not merely a small density tail. Small tail probability alone does not control an unbounded polynomial or derivative integrand. For a constant $c > 0$ the curvature error in (2.9) is the right-hand loss of (6.1) divided by c^2. This conversion is ill-conditioned as $c \downarrow 0$; the zero case uses its own heat identity. If approximate variables violate shape signs or differential rules, those violations require separate residual terms; a small score defect alone does not repair them.

6.2 Coefficient errors as moment errors

Let $Y = (R, K, v, B, j)$ with nonnegative coordinates. If $E_{\mathrm{alg}}(Y) = \sum_\alpha e_\alpha Y^\alpha$ and certified absolute moment bounds give $\mathbb{E}_f|Y^\alpha| \le m_\alpha$, then

$$\eta \le \sum_\alpha |e_\alpha| m_\alpha. \tag{6.2}$$

The proof is the triangle inequality under the integral. A lower Gram bound $Q_i \succeq -\eta_i I$ in a representation $\sum_i g_i z_i^{\mathsf{T}} Q_i z_i$ with $g_i \ge 0$ adds $\sum_i \eta_i \mathbb{E}_f[g_i\|z_i\|^2]$ to the loss. This is the integrated version of the private residual-aware Gram certificate [9]. It is classical polynomial positivity

with explicit moment accounting. The unbounded score variable K prevents silently using the compact-box hypothesis of that private theorem globally. Either certify these moments directly, or combine compact-box bounds with an integrable tail enclosure.

If an exact polynomial in time p on $[a,b]$ is represented after $t = a + \ell s$ as $p = \sum_{k=0}^{d} \alpha_k s^k$, then

$$p = \sum_{j=0}^{d} \beta_j \binom{d}{j} s^j (1-s)^{d-j}, \qquad \beta_j = \sum_{k=0}^{j} \alpha_k \frac{\binom{j}{k}}{\binom{d}{k}}. \tag{6.3}$$

Thus $\min \beta_j \le p \le \max \beta_j$ on the entire interval. The identity follows by expanding s^k in degree-d Bernstein polynomials; nonnegativity and partition of unity prove the bounds. Rational subdivision preserves exactness. Applied to an enclosure for $\gamma^2 M_4$ or e, this can certify the time integral in Corollary 5.4. A polynomial *fit* to a nonpolynomial quantity additionally needs its uniform remainder bound. The corpus's between-knot warning [10] is directly pertinent: finite point checks do not prove the required all-time inequality.

6.3 Ram analysis and synthesis on the actual finite space

Choose a finite polynomial space V_d and basis $\phi_1, \dots, \phi_N$. For specified finite functionals ℓ_i, form $E_{ij} = \ell_i(\phi_j)$ and $m = Ec$ for coefficient vector c. If E is invertible then $c = E^{-1}m$. If rectangular, require full column rank and a specified left inverse on its range. This is the definition used in the private TR250 [8], not a claim that a finite bank recovers an arbitrary Parisi field. For a linear operator $D : V_d \to W$ the coordinate matrix changes by $D^{\mathrm{R}} = E_W D E_V^{-1}$, and a physical mass matrix becomes $M^{\mathrm{R}} = E_V^{-\mathsf{T}} M E_V^{-1}$. Therefore exact polynomial identities and their physical quadratic forms are retained.

The present correspondence is concrete: encode the monomials of (4.5), the derivation (2.7), the Gaussian or density moments in (6.2), and the finite time weights in (5.8). Reconstruction checks algebraic identities on that chosen polynomial space; the separately proved analytic estimates determine whether they represent the PDE. No conditioning gain, compression ratio, or speedup was measured.

TR8's PDE-first equivalence [11] explains why the whole-line heat kernel, terminal trace and nonlinear drift must precede any localized panel. Its bounded-domain coercive inverse is not a theorem about this backward whole-line nonlinear terminal problem. In particular $e^{c\Phi}$ has exponential growth and is not an arbitrary $L^2(\mathbb{R})$ datum, although its Gaussian convolution is finite.

The exact Leibniz rows in the nonlinear-jet source [12] correspond to (5.5): all product derivatives are retained. Its finite recovery theorem requires a nonsingular matrix on a polynomial class. The Parisi field is generally not polynomial, and the collection of its moments is not finitely closed: differentiating a moment introduces higher spatial derivatives. A truncation must retain the omitted term $f^{(N+1)}\partial_{F_N}\Psi$, and coordinate Jacobian terms cannot be replaced by a frozen matrix. These exact passages support a representation and error ledger, not a replacement for the shape, regularity or first-variation arguments.

7 Sharp value stability

Theorem 7.1 (Sharp coefficient stability constant)**.** *For $\gamma, \beta \in \mathcal{U}$, every $t \in [0,1]$ and $x \in \mathbb{R}$,*

$$|\Phi_\beta(t,x) - \Phi_\gamma(t,x)| \le \frac{1}{2} \int_t^1 |\beta(s) - \gamma(s)| \mathrm{d}s. \tag{7.1}$$

The constant $1/2$ *is optimal uniformly in these data. Moreover*

$$|\mathcal{P}(\beta) - \mathcal{P}(\gamma)| \le \frac{1}{2}\int_0^1 |\beta(t) - \gamma(t)| \max\{t, 1-t\}\mathrm{d}t. \tag{7.2}$$

Proof. Repeat the exact difference argument of Section 3 starting at (t, x), with no small parameter. The linearizing diffusion Y has drift $\beta(u_\beta + u_\gamma)/2$, and the exact identity is

$$\Phi_\beta(t,x) - \Phi_\gamma(t,x) = \tfrac{1}{2}\int_t^1 (\beta-\gamma)(s)\mathbb{E}[u_\gamma(s, Y_s)^2]\mathrm{d}s.$$

Its derivation uses the already established $3/2$ tail estimate to remove the terminal difference, so there is no circular use of the improvement. Since $|u_\gamma| \le 1$, this proves (7.1).

For sharpness choose $\gamma = 0$, $\beta = c > 0$, and $t = 0$. Then $\Phi_c(0,x) = c^{-1}\log \mathbb{E}e^{c|x+Z|}$ and $\Phi_0(0,x) = \mathbb{E}|x+Z|$. As $x \to +\infty$, dominated convergence after factoring e^{cx} gives $\mathbb{E}e^{c(|x+Z|-x)} \to \mathbb{E}e^{cZ} = e^{c^2/2}$; domination follows from $||x+Z| - x| \le |Z|$. Also $\mathbb{E}|x+Z| - x \to 0$ by the same bound. Thus $\Phi_c - \Phi_0 \to c/2$ while $\|\beta - \gamma\|_1 = c$.

For the last assertion, the segment $\gamma_\theta = (1-\theta)\gamma + \theta\beta$ lies in $\mathcal{U}$. The first-variation formula gives its derivative for $0 < \theta < 1$. Compact-strip stability followed by dominated convergence in time makes this derivative continuous in θ, with one-sided endpoint limits. Integrating it gives

$$\mathcal{P}(\beta) - \mathcal{P}(\gamma) = \tfrac{1}{2}\int_0^1\int_0^1 (\beta-\gamma)(t)(q_{\gamma_\theta}(t) - t)\mathrm{d}t\mathrm{d}\theta.$$

Since $0 \le q_{\gamma_\theta}(t) \le 1$, the pointwise absolute factor is at most $\max(t, 1-t)$, proving (7.2). □

The improved constant controls the source's coefficient-approximation error directly. It does not improve high-order derivative constants without carrying out the interpolation step and its strip-dependent bounds. For different terminal data an additional terminal supremum error is retained. No finite-precision Parisi solver is claimed.

8 Consequences and remaining boundaries

8.1 Smooth minimizers and the improved density budget

Assuming the source's full-support conclusion, $q = t$ and $F = 1$. Equation (5.6) gives almost everywhere

$$\gamma(t) = N(t)/(2M_3(t)), \qquad M_3 \ge M_2^{3/2} = 1. \tag{8.1}$$

The right side is continuous, and right continuity extends equality to every $t < 1$. At zero, evenness makes $C_0 = 0$, so $\gamma(0) = 0$. Every polynomial moment of the spatial derivatives obeys an integral equation whose right side is a finite sum of other polynomial moments, plus γ times such moments, by (5.5). Once γ and all moments are continuous, these equations make all moments C^1; (8.1) makes γ C^1. Repetition proves C^∞ on each compact strip, including right derivatives at zero. This is an infinite-family bootstrap, not an assertion of finite algebraic closure or time analyticity.

Theorem 5.2 then reduces to

$$2\gamma'(t)M_3(t) = \mathcal{K}(t) \ge \kappa\gamma(t)^2 M_4(t). \tag{8.2}$$

For $t > 0$, full support gives $\gamma(t) > 0$ and $M_4 \ge 1$, so the density is positive. At zero this coercive lower bound is zero and proves no strict positivity there. This improves the source's numerical lower

coefficient but not the interval of validity. As a further conditional consequence, $M_3 \le M_4^{3/4}$ and $M_4 \ge 1$ give $M_4/M_3 \ge 1$, whence $\gamma' \ge (\kappa/2)\gamma^2$ for $t > 0$. For any $0 < s < t < 1$,

$$\frac{1}{\gamma(t)} \le \frac{1}{\gamma(s)} - \frac{\kappa}{2}(t-s). \tag{8.3}$$

In particular finiteness for all $t < 1$ implies $\gamma(s) \le 2/[\kappa(1-s)]$. This coarse upper envelope is not an integrability theorem or a terminal asymptotic: its right side is itself nonintegrable at one, and admissibility remains an independent condition. Indeed monotonicity and admissibility already give $(1-t)\gamma(t) \le \int_t^1 \gamma \to 0$, a stronger asymptotic order statement; the displayed envelope is only an explicit coarse necessary bound.

8.2 Martingale values and scalar Gaussian approximation

The bounded martingale $u(t, X_t)$ has a terminal value U_1 in L^2; saturated consistency gives $\mathbb{E}U_1^2 = 1$ and $|U_1| \le 1$, so $U_1^2 = 1$ almost surely. The uniform terminal estimate and convex difference quotients show $U_1 X_1 = |X_1|$, without needing to exclude $X_1 = 0$. Itô's identity for $\Phi(t, X_t)$, martingality and Fubini then give the source's value formula

$$P_* = \mathbb{E}|X_1| - \int_0^1 t\gamma(t)\mathrm{d}t = \mathbb{E}(U_1 B_1) = \int_0^1 \mathbb{E}A_t \mathrm{d}t.$$

Absolute integrability follows from $\mathbb{E}A_t \le \sqrt{M_2} = 1$.

For completeness, the source's scalar construction first fixes $T < 1$, sets $\Delta = T/N$, and uses

$$Y_j = Y_{j-1} + \sqrt{\Delta} Z_j + \Delta\gamma(t_{j-1})u(t_{j-1}, Y_{j-1}), \qquad g_j = c_j A(t_{j-1}, Y_{j-1}) Z_j,$$

with $c_j = (\mathbb{E}A(t_{j-1}, Y_{j-1})^2)^{-1/2}$. Euler convergence on the fixed strip and $M_2 = 1$ give $c_j \to 1$ uniformly. The g_j are orthonormal martingale differences. An independent uniform random variable V rounds $u(T, Y_N)$ by $F_N = \mathrm{sgn}(u(T, Y_N) + 2V - 1)$, preserving its conditional mean. With $A_j = \mathbb{E}F_N g_j$, $B_i = \mathbb{E}F_N Z_i$, and $A_0 = 0$, Bessel's inequality gives the two ℓ^2 coefficient approximations and $\sum_{j=0}^N A_j B_{j+1} \to \int_0^T \mathbb{E}A_t \mathrm{d}t$. The last term vanishes because $A_N = o(1)$ and $|B_{N+1}| \le 1$. One takes $N \to \infty$ before $T \uparrow 1$. No uniform rate as $T \uparrow 1$ is supplied by this argument.

8.3 External algorithm and literature comparison

The finite-spin optimization result is an application of El Alaoui, Montanari and Sellke [7]: the strictly increasing minimizer supplies their no-overlap-gap hypothesis. Their real-arithmetic, fixed-accuracy algorithm is separate from the scalar construction; neither that theorem nor a faster implementation is proved here.

Current primary literature was checked by public title and topic on October 8, 2026 UTC. Auffinger–Chen establish the zero-temperature variational formula [2]; Auffinger–Chen–Zeng establish infinitely many support points [4], a weaker assertion than absence of every gap. Chen's July 2026 preprint states full relative support and smooth absolute continuity at zero temperature, and its Proposition 4.2 gives a conditional crossing statement, rather than an unconditional positive lower bound on the third derivative [5]. Its conditional formalization explicitly retains analytic assumptions; it is not an independent proof of those inputs. Lopatto's current version is v4, revised October 5, 2026; its public abstract states the positive-temperature interval-support conclusion for every $\beta > 1$ [6]. The pinned manuscript cited v3; the current version is not silently substituted into the pinned proof. These comparisons do not establish historical priority for the local estimates in this report.

Classical ingredients here are Cole–Hopf, Gaussian tilting, parabolic comparison, stochastic control, Itô calculus, BV integration, polynomial positivity, exact linear algebra, and Bernstein enclosure. The private Ram contribution is their explicitly specified finite moment/jet representation and residual bookkeeping. Coordinate changes do not create a new Parisi theorem. The newly derived local statements are Theorems 5.2–6.1 and 7.1, together with the exact finite witness; "newly derived" is relative to this comparison, not a claim of first discovery.

9 Verification, advantages, and future work

The finite checks distinguish three levels. Exact symbolic arithmetic checks polynomial identities, moment derivative algebra, the weighted integration signs, and rational coefficient inequalities. Analytic proofs establish their stated conditional and local consequences. Neither level independently verifies the entire full-support theorem or the external finite-spin algorithm. No sampling experiment, solver timing, or PDE approximation is reported as executed evidence.

The certificate's positive rational remainder makes the core spatial inequality replayable. The BV formulation exposes the precise negative mass term lost by a naive differentiation across a coefficient jump. The weighted formulation turns exact gap exclusion into a test that survives explicitly bounded endpoint and integral errors. The sharp stability estimate improves a genuine source bound with a limiting example establishing optimality of the constant.

The disadvantages are equally specific. Compact-strip constants need not remain useful near time one. Density-score errors must be measured in the actual integration law, and a numerical density requires its own law-approximation certificate before giving a PDE conclusion. The Bernstein route applies directly only to exact polynomials or proved polynomial enclosures. The moment bank must be unisolvent and well conditioned; positivity of its weights alone does not suffice. The restricted cone optimum does not imply sharpness of the analytic curvature inequality.

Useful next work is to derive computable compact-strip derivative constants, certify positive m_4 and finite V_3 on concrete candidate intervals, and combine a rigorous forward-density enclosure with (6.1). One may also enlarge the divergence and square ansatz while retaining an exact dual or an explicit residual budget. Extensions to nonzero field or nonconstant ξ'' require new shape identities and drift calculations: neither is implied by these local gains. The $C3$ assessment credits the realized estimates and replayable algebra, not the inherited support assertion, future numerics, or an unmeasured change in computational cost.

A Exact optimum in a restricted coefficient cone

Fix $Q, T, P_0, \dots, P_4$ as in (4.1)–(4.2) and vary only $(p_0, \dots, p_4, x, y, m, n)$, with $p_i \geq 0$, $x \geq 0$. Use F of the form (4.3). After the exact z, L_s cancellations and $b = B + 2v$, require every remainder monomial coefficient to be nonnegative. Maximizing the coefficient of v^4 is a finite rational linear program. Its attained optimum is

$$\delta_{\mathrm{cone}} = 116143777877/57728642400000. \tag{A.1}$$

One attaining vector in the displayed order is

$$\begin{aligned} p_0 &= 2665655747849/76971523200000, & p_1 &= 1/16, \\ p_2 &= 1369/32500, & p_3 &= 2187001362679/962144040000, \\ p_4 &= 32747756957/513143488000, & x &= 14275776508471/15394304640000, \\ y &= 39540945483719/57728642400000, & m &= -8081913089/3078860928, \\ n &= -2754281365999/1282858720000. \end{aligned}$$

The nine active coefficient rows in order are

$$Bv^3, B^2Kv, BKv^2, BRjv, B^2Rv, Rjv^2, Rv^3, KRv^2, R^2v^2.$$

Here the fifth active row may have zero dual weight. If the coefficient vector is $r(u) = r_0 - Mu$, and t is its v^4 row, the rational vector on those nine rows is

$$\begin{aligned} (\lambda_1, \dots, \lambda_5) &= (-64512/237215, -4608/47443, -4608/237215, -288/47443, 0), \\ (\lambda_6, \dots, \lambda_9) &= (-576/237215, -1/5, -192/237215, -47379/237215). \end{aligned}$$

satisfies $M_I^{\mathsf{T}}\lambda = M_t^{\mathsf{T}}$, $\lambda \le 0$ and $r_{0,t} - \lambda^{\mathsf{T}} r_{0,I} = \delta_{\text{cone}}$. Therefore

$$r_t(u) = \delta_{\text{cone}} + \lambda^{\mathsf{T}} r_I(u) \le \delta_{\text{cone}}$$

for every feasible vector. At the attaining vector the active rows vanish and all other coefficients are nonnegative. These equalities, including the nonnegative 13-term remainder, were checked over the rationals; numerical LP discovery is not the truth criterion. The strict-positive 22-term witness in the main text is easier to perturb and gives the clean 1/500 lower bound. No optimality is asserted if the squares, Q, T, monomial cone or b-dependent correction are changed.

B Exact read scope and private correspondences

All source files below are under

C:/Users/mural/OneDrive/Documents/OpenAI-722-math/_sources/openai-math/preprints/Full-support-of-the-zero-temperature-Sherrington-Kirkpatrick-order-parameter-September-27-2026/build/

File	Substantive scope read
`main.tex`	Model, full-support theorem, prior-work comparison, proof structure and every input declaration.
`regularity.tex`	All smoothing, approximation, diffusion and constant-interval statements and proofs.
`signs.tex`	All differentiated-tail, maximum-principle, backward, forward, ratio, jump, quotient and IBP arguments.
`certificate.tex`	All rational constants, derivation, 22 coefficients, identity proof and analytic coercivity hypotheses.
`curvature.tex`	Entire finite-step to integrable-coefficient transfer and zero-coefficient branch.

File	Substantive scope read
variation.tex	All admissible directions, compact mass removal, contact, terminal layer and gap-component proof.
order-parameter-regularity.tex	Entire moment-ratio bootstrap and fixed-time flattening proof.
consequences.tex	All terminal-martingale, Euler, normalization, rounding, Bessel and shifted-product proofs.
algorithm-application.tex	All external-theorem, covariance, probability and complexity qualifications.
gap-convexity.tex	Complete figure and mathematical caption, not used as proof.
glyph-mappings.tex	All rendering definitions; no mathematical claim.
references.bib	Complete bibliography; public references checked separately where used.

Private passages were read directly at the following exact paths and sections. Routing maps were not evidence for these statements.

1. C:/Users/mural/OneDrive/Documents/RamFoundationTRs/TR-rev-1-old-tr250/TR-rev-1-old-tr250.tex: "Spaces, operator domains, and Ram coordinates", especially "What a Ram representation means in this report", eq:moments; "Ram moment coordinates that retain the complex", thm:transfer; and "A concrete polynomial Ram construction". The application here preserves a finite polynomial identity and its physical moment functional, not a continuous differential complex claimed for the Parisi field.

2. C:/Users/mural/OneDrive/Documents/MyDocuments2026/res-rt/res-math/TR-item-047-astra.tex: "Algebraic and Moment Reformulation", "Ram Analysis and Synthesis Panels", "Global Optimality Sandwich", theorem "Replayable global gap certificate", eq:gap; and "Residual-Aware SOS Lower Bounds", sec:soserror047, theorem "Inexact Gram and identity certificate", eq:gramerror047. The compact-box hypothesis is replaced here only when explicit absolute moments are supplied.

3. C:/Users/mural/OneDrive/Documents/RamCalBooks/volumes/volume16/chapters/ch05d-clearance.tex: "Between-knot constraints are continuous-time conditions", "An independent dynamics defect", "Polynomial enclosures in Bernstein form", v16c05:powerbernstein, v16c05:conversionproof, v16c05:hull, and "Subdivision without introducing approximation". Their direct analogue is an all-time enclosure of the integrand in the mass bound.

4. C:/Users/mural/OneDrive/Documents/RamNonlin-IE-PDE-astra/TR8-lin-non-lin/TR8-lin-non-lin.tex: "PDE-first scope and notation", sec:elliptic, thm:coercive, eq:operator-domain, eq:compatibility, thm:green-equivalence, "Parabolic initial-boundary problems and Duhamel's formula", thm:heat-equivalence, and sec:semilinear. These passages identify exactly which hypotheses a kernel or localized representation does not provide.

5. C:/Users/mural/OneDrive/Documents/RamNonlinearIE-PDE/TR1-lin-nonlin-rev2/sec-03-jets.tex: "The exact Ram-moment jet with remainder", differentiated rows thm:t1-leibniz-rows, recovery thm:t1-recovery, degree closure thm:t1-degree-closure, and nonlinear map thm:t1-nonlinear-jetmap, including eq:t1-jet-jacobian and

`eq:t1-jet-closure-defect`. The nonlinear theorem's exactness on a polynomial source class does not give a finite closed Parisi moment system.

The private material stays local. Public literature searches used only public mathematical titles and topics; none of the private passages or report text was submitted in those queries.

References

[1] OpenAI. *Full support of the zero-temperature Sherrington–Kirkpatrick order parameter*. September 27, 2026. Catalogue M565, pinned commit stated in Section 1. https://github.com/openai/math/blob/adc7f1241b42e322a6451854ab7e4b4c146bf78a/preprints/Full-support-of-the-zero-temperature-Sherrington-Kirkpatrick-order-parameter-September-27-2026/main.pdf.

[2] A. Auffinger and W.-K. Chen. Parisi formula for the ground state energy in the mixed p-spin model. *Ann. Probab.* 45 (2017), 4617–4631. https://arxiv.org/abs/1606.05335v2.

[3] W.-K. Chen, M. Handschy and G. Lerman. On the energy landscape of the mixed even p-spin model. *Probab. Theory Relat. Fields* 171 (2018), 53–95. https://arxiv.org/abs/1609.04368v4.

[4] A. Auffinger, W.-K. Chen and Q. Zeng. The SK model is infinite step replica symmetry breaking at zero temperature. *Comm. Pure Appl. Math.* 73 (2020), 921–943. https://arxiv.org/abs/1703.06872v2.

[5] H.-B. Chen. *FRSB in the SK spin glass: convergence to full-interval support at zero temperature*. July 20, 2026, v1. https://arxiv.org/html/2607.18032v1. The manuscript includes an explicit AI-authorship disclosure and conditional formalization boundary.

[6] P. Lopatto. *Full replica symmetry breaking in the Sherrington–Kirkpatrick model*. v4, October 5, 2026. https://arxiv.org/abs/2607.11756v4. The pinned source cites v3.

[7] A. El Alaoui, A. Montanari and M. Sellke. Optimization of mean-field spin glasses. *Ann. Probab.* 49 (2021), 2922–2960. https://arxiv.org/abs/2001.00904v1.

[8] M. Subbarao. Private TR250 revision, `TR-rev-1-old-tr250.tex`; exact path and passages in Appendix B. Confidential local corpus.

[9] M. Subbarao. Private mathematical report `TR-item-047-astra.tex`; exact path and passages in Appendix B. Confidential local corpus.

[10] M. Subbarao. *Ram Calculus Books*, volume 16, `ch05d-clearance.tex`; exact path and passages in Appendix B. Confidential local corpus.

[11] M. Subbarao. Private nonlinear/PDE report 8, `TR8-lin-non-lin.tex`; exact path and passages in Appendix B. Confidential local corpus.

[12] M. Subbarao. Private nonlinear integral-equation report 1, revision 2, `sec-03-jets.tex`; exact path and passages in Appendix B. Confidential local corpus.